Part A: Conceptual Understanding (Theory)

Classification Concepts – Brief Answers with Formulas and Graphs

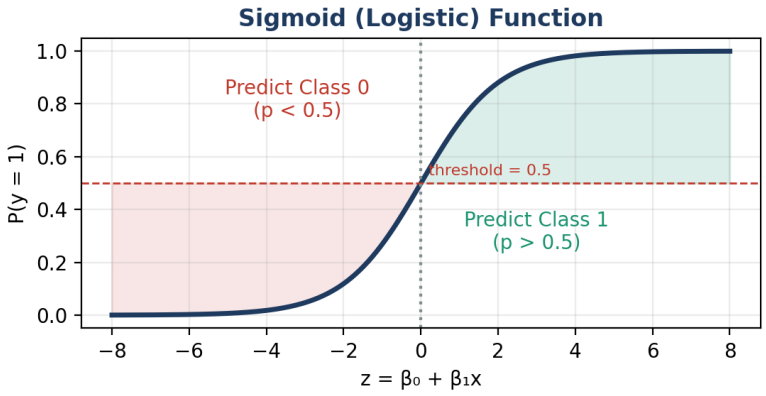
1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised algorithm that predicts the **probability** of a class (0 or 1) by passing a linear combination of features through the **sigmoid function**, which squashes any value into the range 0 to 1. A threshold (usually 0.5) converts this probability into a class label.

It is suitable for classification because its output is a bounded probability (unlike linear regression), it is fast and interpretable, and it is trained with log-loss, which suits categorical targets.

$$P(y = 1 \mid x) = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \ \ldots \ + \beta_n x_n$$

$$\text{Log-loss} = -\frac{1}{N}\sum [y \log \hat{p} + (1 - y)\log(1 - \hat{p})]$$

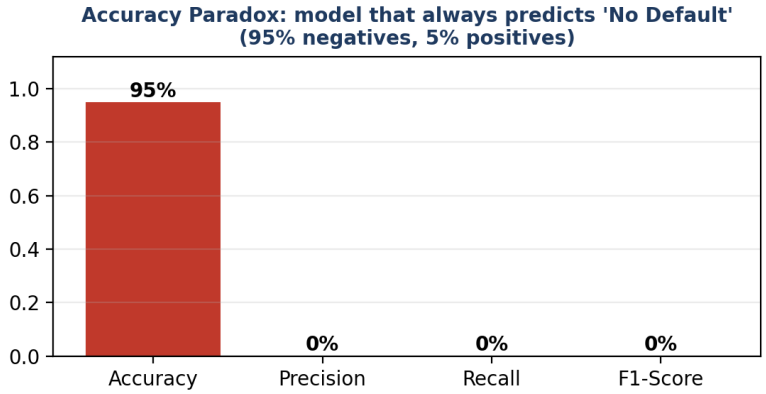


2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) measure how well predicted labels match actual labels, all derived from the confusion matrix (TP, TN, FP, FN).

Accuracy alone is insufficient because on imbalanced data a model can score very high by always predicting the majority class while missing every positive case. It also treats all errors as equal, though a missed defaulter costs far more than a false alarm.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

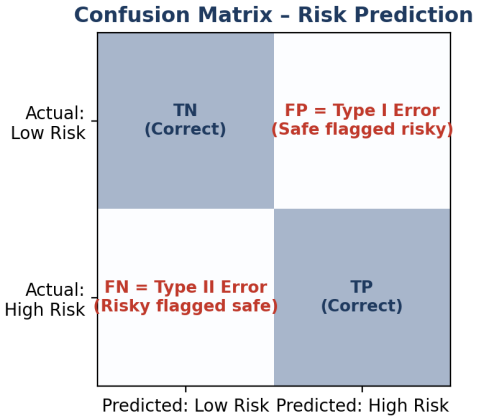


3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive): a safe/low-risk customer is wrongly predicted as high risk, e.g. a good applicant is rejected a loan. The cost is lost business.

Type-II Error (False Negative): a risky customer is wrongly predicted as safe, e.g. a likely defaulter is approved. The cost is financial loss, usually more serious in risk prediction.

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true}) = \frac{FP}{FP + TN} \qquad \beta = P(\text{Accept } H_0 \mid H_0 \text{ false}) = \frac{FN}{FN + TP}$$



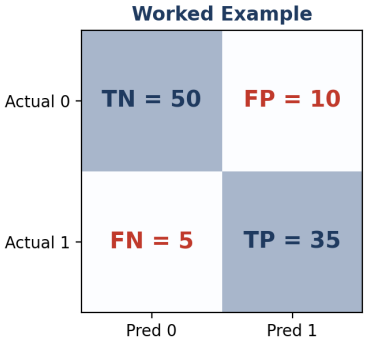
4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision: of all predicted positives, how many are truly positive. **Recall / TPR:** of all actual positives, how many were caught (sensitivity). **F1-Score:** harmonic mean of Precision and Recall, balancing both. **FPR:** of all actual negatives, how many were wrongly flagged positive.

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall (TPR)} = \frac{TP}{TP + FN}$$

$$\text{F1} = \frac{2 \cdot P \cdot R}{P + R} \qquad \text{FPR} = \frac{FP}{FP + TN}$$

Worked example (TP=35, FP=10, FN=5, TN=50): Precision = 35/45 = **0.78**, Recall = TPR = 35/40 = **0.875**, F1 = **0.82**, FPR = 10/60 = **0.17**.

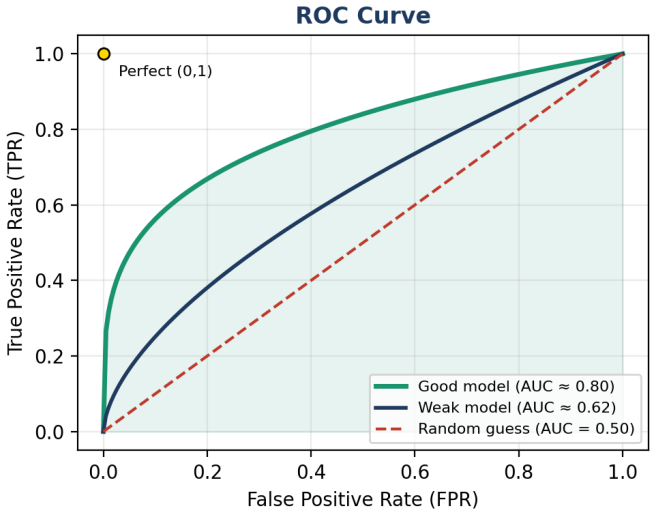


5. What is AUC-ROC and how does it help in evaluating classifiers?

The **ROC curve** plots TPR against FPR at every classification threshold. **AUC** is the area under this curve: 1.0 means perfect separation, 0.5 means random guessing.

It helps because it is **threshold-independent**, shows the trade-off between catching positives and raising false alarms, and lets you compare different models with a single number.

$$\text{AUC} = \int_0^1 \text{TPR}(t)\, d(\text{FPR}(t))$$



6. Why does imbalanced data create problems in classification models?

When one class dominates (e.g. 95% vs 5%), the model learns mostly from the majority class and becomes **biased** toward it. It gets high accuracy but very low recall on the minority class, which is usually the one that matters (fraud, default, disease).

Fixes: resampling (SMOTE, undersampling), class weights, threshold tuning, and evaluating with Precision, Recall, F1 and AUC-ROC instead of accuracy.

